

Assignment: Building a Mini Kinyarwanda Voice Assistant

Course: Intelligent Robotics

Instructor: Gabriel Baziramwabo

Assigned: April 23, 2025

Due: April 30, 2025 (11:59 PM)

Duration: 1 week

Level: Advanced

Why This Matters in Intelligent Robotics

In modern humanoid robots, voice interaction is essential. These robots must:

- Hear and transcribe human speech (ASR),
- Understand the question (NLP),
- Respond using speech (TTS or actions).

In this assignment, you'll simulate this exact process but in Kinyarwanda.

You are essentially giving a humanoid robot:

- Ears: KinyaWhisper (Speech to Text),
- Brain: NLP logic to understand questions,
- Mouth: TTS to reply using voice.

This project shows how intelligent robots can communicate in local languages, empowering them to assist Rwandan communities.

Objectives

- Build a simple Kinyarwanda voice assistant.
- Convert speech to text using KinyaWhisper.
- Match questions to answers using basic NLP.

- Use Text-to-Speech to speak answers back.

Learning Outcomes

1. Apply ASR models in practical robotics use cases.
2. Simulate how robots interpret and respond to speech.
3. Build a functioning prototype using GitHub.
4. Gain skills in ASR, NLP, TTS the core of voice-enabled robotics.

Tools You Can Use

- KinyaWhisper (for ASR)
- Python 3
- Coqui TTS or gTTS (for voice output)
- Git + GitHub (required)
- (Optional) Gradio or Streamlit for GUI

What To Do

1. Speech Recognition (ASR)
 - Record or receive audio in Kinyarwanda.
 - Use KinyaWhisper to transcribe it into text.

2. NLP Matching

- Create a dictionary of at least 5 Kinyarwanda QA pairs.

3. Text-to-Speech (TTS)

- Use TTS tools to speak the answers back.

What To Submit

1. Push your full code and data to a public GitHub repository.
2. Ensure the repo contains:
 - Python files or notebooks,
 - A README.md explaining the project and how to run it,
 - At least 5 audio files and their transcriptions,
 - NLP mapping and matched answers,
 - Spoken answer outputs.
3. Submit your GitHub repo URL via the Google Form provided .

Google Form for Submission

Submit Your GitHub URL Here: <https://forms.gle/ZpkypNATrkfeyCLO7>

Evaluation Criteria (100 Marks)

Criteria	Description	Marks
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ASR Accuracy	Proper transcription of Kinyarwanda speech	25
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NLP Matching	Correct question-to-answer logic	25
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TTS Output	Clear, intelligible spoken answers	20
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Code Quality	Structured, commented, readable code	15
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GitHub Usage	Proper commits, README, organized repo	10
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Submission	Link submitted on time via form	5
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